

VGIC Informal Feedback on Dominion DSM-XIV Proposal

Dominion's questions are listed below in **bold**; VGIC's responses are listed below each bolded question.

Residential Battery Questions

What considerations or concerns should the Company address to responsibly allow participants to export power from their batteries to the grid?

Dominion should ensure it provides broad eligibility for residential battery systems, including both stationary energy storage systems and bidirectional EV charging systems. Notably, the available grid-parallel BESS and grid-parallel DC bidirectional EV charging systems all comply with the same IEEE 1547-2018 standard via UL 1741 SB certification. These standards are critical to ensure grid safety and interoperability, and automakers and charging manufacturers have designed their solutions to ensure compliance with these specifications. In multiple cases, the DC bidirectional EV charging systems available to customers use the very same inverter models used in stationary BESS. VGIC strongly encourages Dominion to deem DC bidirectional EV charging systems eligible for its residential battery program and subject these systems to the same underlying IEEE 1547-2018 and UL 1741 SB requirements expected of modern stationary BESS systems, and we offer ourselves as a resource to discuss additional technical details regarding this eligibility consideration.

As a participant considering injecting power back into the grid, how should the Company weigh the value for a participant receiving net metering credits for exporting excess energy versus earning battery performance incentives that reflect a participants system's contribution to grid reliability and responsiveness?

Device-level measurement (e.g., inverter readings) of battery contributions can serve as the basis for any performance-based compensation provided to the customer as part of the DSM program. If a customer also has a NEM system, this device-level battery export data can be subtracted from any kWh exports detected at the customer meter, such that the remaining value serves as the basis for NEM credits. This ensures NEM credits are only awarded to kWh generated from NEM-eligible resources (i.e., Renewable energy resources as defined in Virginia Code § 56-576). Notably, stationary residential batteries and bidirectional charging systems alike come with embedded measurement capabilities that can support this use case. This approach also retains flexibility for battery export performance payments to customers to be made on a \$/kWh, \$/kW, and/or other basis.

Furthermore, this would not require making any changes to a customer's NEM tariff/agreement.

Alternatively, Dominion can use existing, trusted modeling tools to assess how much solar a given customer is expected to produce over a given period (e.g., billing period or year), and that kWh value can then serve as a cap for the customer's NEM credits. Using this approach, the total kWh exported at the meter can exceed the production from NEM-eligible resources by any amount, for example due to battery exports, V2G exports, or export from any other resource, but the customer's NEM credits would not benefit from these exports coming from non-NEM resources. For compensating the non-NEM resources, device-level measurement can be used in this case, as well.

Both of these approaches have either been adopted or are pending approval in other jurisdictions as a method to ensure "NEM integrity," ensuring that rewarding battery export capability does not mean paying customers NEM rates for non-renewable energy.

As the Company considers an incentive structure that scales based on the different benefits participating customers provide, would stakeholders prefer a simple and easily understandable incentive model based off performance, or a more advanced structure that reflects differentiated value contributions?

Different customers likely have different preferences when it comes to incentive structures. If Dominion intends to focus its forthcoming DSM implementation on launching a large-scale, mass-market opportunity for a range of customers, VGIC recommends starting with simple and easily understandable incentive models, so long as they are sufficiently attractive to prospective customers. Dominion should provide a simple model of customer dispatch, based on reasonable inputs and assumptions for residential batteries and/or bidirectional EV charging systems, that demonstrates a payback period under 5-7 years, depending on the technology type.

Alternatively, if Dominion intends to pursue its Residential Battery offering as a small-scale pilot and learning opportunity, VGIC recommends conducting A-B testing with compensation models. For example, one customer group can be offered a flat \$/month incentive, while the other customer group is offered a time-of-use export credit. This allows Dominion to understand, first hand, both the customer response to different incentive models, and the utility's own internal operations and planning teams interaction with different incentive structures.

What are your expectations for how the Company should measure battery performance directly at the device level, and what data or standards should be prioritized?

VGIC reiterates its recommendation that Dominion measure battery performance directly at the device level, leveraging the embedded capabilities of these customer-sited solutions. As it relates to electric vehicle resources, we recommend leveraging the existing NIST Handbook 44.0 accuracy standard currently used across the country to ensure accurate measurement of EV charging equipment. EV charging equipment manufacturers already work to meet this accuracy standard, and it is used in some cases outside of Virginia to measure kWh for customer billing purposes.

What finance options or incentive structures are customers looking for to participate via a third party?

Based on VGIC's understanding as it relates to bidirectional charging systems, customers seek sufficient (a) upfront incentives to help offset the capital expenditures of installing a bidirectional EV charging system and (b) fair, full value stack (based on avoided costs) compensation mechanisms that help customers recover any upfront costs not offset through upfront incentives and a reasonable return on investment for similar investment assets. Balancing upfront incentives and ongoing compensation supports continued participation.

For customers participating through third-party-owned battery systems, would you prefer to receive program payments directly, or would you rather have those payments assigned to your third-party owner or aggregator to manage compensation and services on your behalf?

VGIC encourages Dominion to remain flexible regarding ownership and compensation models. The program should be flexible enough to accommodate a variety of innovative business models that maximize customer participation and value to the electric system.

How should the Company structure participation options for customers who prefer to engage through third-party aggregators, and what roles or responsibilities should be defined?

If Dominion chooses to offer both direct enrollment and enrollment via a third-party aggregator, VGIC recommends that third-party providers not be placed at a disadvantage relative to direct enrollment via Dominion. For example, the customer should be eligible for equal upfront incentives and/or performance-based compensation regardless of which enrollment pathway they choose. Additionally, qualified aggregators should be afforded the opportunity to access relevant customer data, if given, the customer provides authorization to do so, to support a customer's decision making, site design, and technology selection processes. Again, VGIC reiterates the need to be flexible in program design to allow for business model innovation.

As the Company designs this pilot, how important is it to stakeholders that customers be allowed to participate in multiple demand response programs simultaneously, and what concerns or benefits do you foresee if participation is limited to one program at a time?

Participation across multiple demand response programs is appropriate if the programs can (a) each reward customers only for the differentiated grid services and appropriate values unlocked by that program and (b) avoid double-counting contributions in the measurement and evaluation of the program, so as to prevent overestimating contributions to cost-effectiveness testing and/or grid planning processes. If a given DSM (or VPP) program offers the full avoided cost value stack to customers – including, but not limited to, avoided generation capacity, avoided generation energy, avoided transmission capacity, avoided distribution capacity, avoided ancillary service costs, avoided electrical line losses, voltage support, black start, and avoided environmental compliance costs – then the customer should not be permitted to participate in other programs.

Aggregator Access

Do you have capabilities to support locational dispatch for aggregated DERs?

VGIC represents 42 member companies, including automakers (Ford, GM, Honda, Nissan, Tesla, Rivian, Stellantis, Toyota, BMW, & Mercedes-Benz), third-party aggregators (WeaveGrid, ChargeScape, ev.energy, EnergyHub, & Kaluza), EV charging manufacturers (ChargePoint, Emporia, Tesla, Rivian, & dcbel), EV charging site developers (The Mobility House, GridBeyond, Gravity, Critical Loop, Tesla, Rivian, Ford, & GM), and other providers in this space. Given discussion with our diverse collection of EV-focused members, as well as

other partners in the VGI space, we can state with confidence that aggregators do have the capabilities to support locational dispatch for aggregated DERs. In some cases, this can be as granular as local neighborhood transformers. These companies recognize the fact that in many circumstances, distribution constraints represent a key barrier to electrification and an opportunity to deploy solutions to address these constraints and create value through deferred or avoided distribution infrastructure investments.

What would you require from DEV to support locational dispatch?

VGIC offers no response at this time, but may choose to weigh in on this question in further stakeholder engagement opportunities with Dominion.

More granular data on the distribution system will be required to determine optimal strategies for locational DER dispatch. Many jurisdictions require utilities to develop detailed distribution system plans that can consider load flexibility and distribution exports alongside traditional investments to achieve least cost solutions and rate payer value.

What resources should be permitted to participate in the program and why?

VGIC respectfully urges Dominion to deem residential bidirectional EV charging resources eligible for the program. These systems are increasingly available to customers, with capable vehicle offerings currently available from Ford, GM, Tesla, Kia, and soon to be released from Honda, Nissan, Mercedes-Benz, BMW, Stellantis, Hyundai, Rivian, Lucid, and Toyota. Additionally, there is corresponding bidirectional charging equipment from Wallbox, dcbel, Ford, GM, and Tesla, and soon to be released from ChargePoint, Emporia, Autel, Enphase, and several others. Thousands of these vehicles have been deployed across the U.S., with grid-parallel interconnected and net exporting sites online in Maryland and, in just a matter of weeks, in Massachusetts and California. While there are more bidirectional capable vehicles, certified chargers, published standards, available compensation mechanisms, and appropriate interconnection rules than there have ever been, the most significant reason VGIC believes these systems should be eligible to participate is the customer interest. In facilitating program implementation in other states, VGIC can attest to the engagement and customer demand for new V2G programs from a small but quickly growing subset of EV drivers.

Should any technologies be excluded and why?

VGIC offers no response at this time but may choose to weigh in on this question in further stakeholder engagement opportunities with Dominion.

What payment structure would you recommend for aggregators?

- **Should incentives be paid directly to aggregators, or routed through a program vendor implementor**
- VGIC encourages flexibility as the market evolves and new business models emerge to meet customer interest.
- **What models (e.g., performance-based, fixed fee, tiered) would best support participation and impact?**
- VGIC recommends a compensation model that balances flat, upfront incentives to promote technology adoption and customer enrollment with ongoing, performance-based compensation tied to grid services value. As detailed in VGIC's response to question 3 above, implementation intended to achieve large-scale enrollment as quickly as possible should err for simpler mechanisms and consider introducing more complex models as interest grows and the market evolves. However, if Dominion aims to implement a limited scale pilot focused on developing valuable learnings that can inform later program design, VGIC recommends Dominion experiment with different incentive designs to gauge customer response and relative implementation difficulty for its internal teams.

What requirements should be in place to measure aggregator performance for settlement purposes?

- **What metrics or data should be used?**
- VGIC offers no response at this time, but may choose to weigh in on this question in further stakeholder engagement opportunities with Dominion.
- **How frequently should performance be reviewed?**
- VGIC recommends settlement be provided each billing cycle. Annual settlement should be avoided, as it can be challenging for customers to link their participation and resulting financial gain.

What controls should be implemented to prevent dual participation?

Aggregators should offer Dominion the relevant customer data for verification that a customer is not enrolled in (a) the same program with another aggregator or (b) another program in which cross-enrollment is prohibited. Clearly defining the grid services and the associated performance can serve to provide clarity and avoid double compensation.

What types of data should be shared with aggregators to support program delivery and transparency?

Assuming a customer has provided authorization, aggregators should be afforded access to certain customer data, which may include that customer's meter data, to facilitate customer engagement. This access may be necessary to provide before a customer formally enrolls in a given program, as it may be an input into the customer's choice of program and/or technology investment. Furthermore, advanced access to this data is necessary to forecast revenue from program participation and thus support leasing or other innovative financial arrangements to maximize customer engagement.

What types of penalties or consequences should be included for non-compliance or underperformance?

VGIC offers no response at this time, but may choose to weigh in on this question in further stakeholder engagement opportunities with Dominion.

Are there any additional considerations the Company should take into account regarding aggregator access to utility programs that have not been addressed above?

VGIC offers no response at this time, but may choose to weigh in on this question in further stakeholder engagement opportunities with Dominion.

Residential Managed Charging (DR) Program

What type of incentive structure do you believe would entice our residential customers to participate in a Res. Managed Charging (DR) program? Please include any general thoughts on incentives and eligibility for those on TOU rates versus non-TOU rates.

Customers would likely be enticed to engage in a residential managed charging program that offers various options to enroll (including through either the vehicle or charger).

Regarding incentive structures, VGIC recommends a balance of flat enrollment/participation incentives and performance-based incentives to ensure customers are attracted to the program and that they continue to provide Dominion with robust DSM tools.